MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5726

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin diode, miniature

Outline --- 5-1 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode -- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	1k	2a	h	h	2k	sd	1a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Epp/a	epx	Ehk	R1	C1	Io/a	ib/a	i surge/a	TE	Alt
Unit:	V	Vac	v	v	Ohms	µF	mAdc	ma	ma	°C	ft
Maximum	6.9	---	360 (see note 1)	360	---	---	9.0	54	320	+165	Note 2
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	165	---	---	11,000	8	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	275	325	mA
1336	Heater-cathode leakage	See note 3	0.4	II	Ihk	---	5	µAdc
1353	Operation of rectifiers	See notes 4 and 5	0.4	II	Io	16.0	20.0	mAdc
1231	Emission (1)	Eb = 10 Vdc (see notes 3 and 5)	0.4	II	Is	40	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	E(a-all)=-300 Vdc E(sd-all)= -300 Vdc (see note 3)	2.5	S3	R	200	---	Meg
1231	Emission (2)	See notes 3 and 6	2.5	S3	ΔIs Ef	---	15	%
1256	Electrode current (anode)	Ebb = 0; Rp = 40,000 ohms (see note 3)	2.5	S3	Ib	2	20	µAdc
---	Electrode current (anode) (difference between diodes)	Ebb = 0; Rp = 40,000 ohms (see note 3)	2.5	S3	Ib	---	5	µAdc
---	Hum	Ef = 7.0 Vac (see note 7)	2.5	S3	E	---	10	mVac
1331	Direct-interelectrode capacitance	Shield No. 316 all measurements						
		1a to 2a			C	---	0.026	pF
		1a to h + 1k + shields			C	2.4	4.0	pF
		2a to h + 2k + shields	6.5	Code E	C	2.4	4.0	pF
		1k to h + 1a + shields			C	3.1	4.7	pF
		2k to h + 2a + shields			C	3.1	4.7	pF
1031	High-frequency vibration	No voltages	6.5	Code E	---	---	---	---
1041	Shock	675 G; Epp/a = 0	6.5	See note 8	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1031	Vibration-fatigue		6.5	See note 8	---	---	---	---
---	Shock and vibration-fatigue test end points:							
1336	Heater-cathode leakage		---	---	Ihk	---	15	µAdc
1353	Operation of rectifiers		---	---	Io	14	---	mAdc
1121	Base strain		---	---	---	---	---	---
2126	Envelope strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = 135 Vdc; Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ihk	---	20	µAdc
1501	Intermittent life test	TE = +165°C (min) (see notes 9 and 10) Group E	---	---	---	---
---	Intermittent life-test end points (500 hours):	See note 11				
---	Inoperatives		---	---	---	---
1301	Heater current		If	275	330	mA
1231	Emission (1)		Is	35	---	mAdc
1231	Emission (2)		ΔIs	---	20	%
			Ef			
1336	Heater-cathode leakage		Ihk	---	10	µAdc
1211	Insulation of electrodes		R	100	---	Meg
---	Total defectives		---	---	---	---
---	Intermittent life-test end points (1,000 hours):	See note 11				
---	Inoperatives		---	---	---	---
1301	Heater current		If	275	333	mA
1231	Emission (1)		Is	30	---	mAdc
1231	Emission (2)		ΔIs	---	25	%
			Ef			
1336	Heater-cathode leakage		Ihk	---	10	µAdc
1211	Insulation of electrodes		R	75	---	Meg
---	Total defectives		---	---	---	---

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NOTES:

1. The maximum voltage appearing between any pair of pins shall be no greater than the peak inverse anode voltage rating.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. Test each unit separately.
4. In a full-wave circuit, adjust Zp/a so that a bogey tube gives Io = 18 mAdc. A bogey tube has a voltage drop of Etd = 10 Vdc at Is = 60 mAdc per anode. Ehk = Eo + 117 Vac.
5. This test shall be performed at the conclusion of the holding period.
6. Test at Eb = 7.0 Vac with readings at Ef = 6.3 V and Ef = 5.7 V.
7. The test circuit to be used is shown in figure 1.

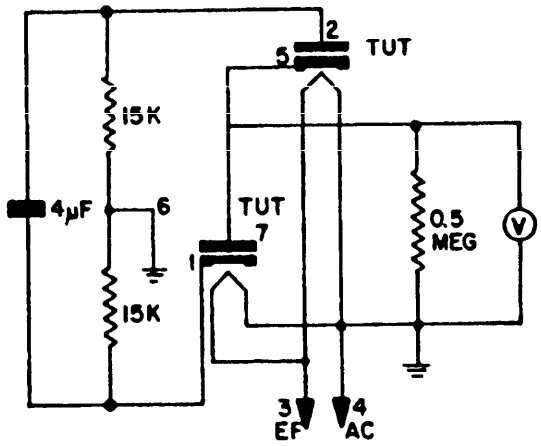


FIGURE 1. Test circuit.

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NOTES: -Continued

8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

9. In a full-wave life-test circuit, the values specified for R1 and C1 may be considered as approximate and shall be adjusted initially to give not less than Io = 18 mAdc and ib = 50 ma per anode with a bogey tube. Ehk = Eo + 117 Vac.

10. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having Etd = 10 Vdc (±5 percent) at Is = 60 mAdc per anode, under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

11. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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